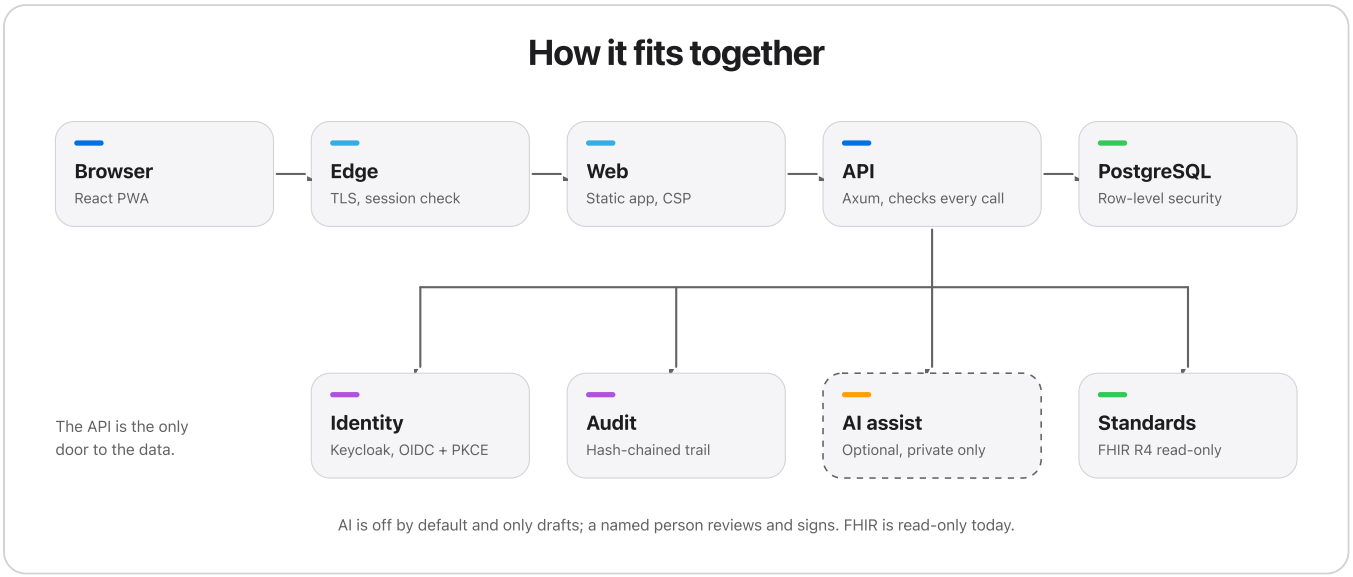
ARCHITECTURE AND DEPLOYMENT

One service, one database, nothing else to run.

A modular monolith: a Rust API in front of PostgreSQL, and a React progressive web app.



Deployment options

SHAPE	USE FOR	VERIFIED HERE
Docker Compose	One host, demos, small clinics	Built and run in a FluxVM Debian VM; smoke and browser tests
Helm chart	Kubernetes, optionally with CloudNativePG	Linted and rendered with kubeconform; installed on k3s
k3s with identity	Keycloak sign-in and an Envoy Gateway edge	Sign-in, session check and sign-out browser-tested; single node, self-signed certificate

Requirements and limits

- PostgreSQL 14 to 18 with `btree_gist` and `pg_trgm`; the same two container images everywhere.
- Integrations: HL7 v2 and ASTM intake, read-only FHIR R4, webhooks. FluxVM is verified live; the Nuvora wire contract was checked against a real Nuvora using its offline demo provider; Netra is tested against a stand-in.
- Not yet: load testing, scheduled backup automation (the installer has backup and restore commands), multi-replica behaviour of the Helm chart.